

Changing standards and technologies

Today, if we talk about the standards and the technologies associated with wireless communication, we will notice that there are fewer standards than technologies. Of course, there is a continuous research and innovation in baseband technologies to make them transmit/receive data faster and securely in noisy environments. Of all the standards in wireless communication, IEEE has become the most popular over the years (apart from 3GPP) due to its broad coverage of wireless technologies for different applications. IEEE standards like Wi-Fi and WiMAX provide high data transfer rates (1Tbps to 150Mbps), depending on the availability of the network. At the other end of the spectrum, a whole range of technologies under the 802.15 umbrella cover specialised applications such as wireless personal area networks, body area networks and industrial wireless communications.

Purkayastha says, "There is a considerable amount of research going on in RF side to develop software-defined radios with tunable RF support for better long-range wireless communication (4G/5G), personal area network and sensor area networks (802.15 umbrella). Then there is research going on to increase the interference cancellation techniques associated with the technologies." Most of the wireless communication technologies developed in the past are meant for use in applications that required clients to download data from the network most of the time. However, new applications like SmartGrid and SmartCities require sensors to upload data back to the grid all the time. Therefore wireless interfaces need to be designed to be more uplink efficient.

Wireless Gigabit (WiGig). WiGig is based on the more advanced 802.11ad standard and was named so by Wi-Fi Alliance. It allows a super-fast video streaming with a data transfer speed of up to 7Gbps. WiGig is based on the triband frequencies (2.4GHz, 5GHz and 60GHz) and has a network operating range of more or less nine metres only, so the triband chipsets can be considered to be advantageous as one can seamlessly switch between the 802.11n, 802.11ac and 802.11ad (WiGig) wireless standards.

Bluetooth Smart. Although it is the same old Bluetooth technology without much change, Bluetooth Smart uses less power as it does not send out signal all the time. This technology is better known as BLE and is widely used in almost every application, from home automation to medical devices and from mobile payments to retail geo-fencing. Companies like Apple and PayPal are working to build applications based on this technology to push phone upgrade messages and to implement a voice recognition technology.

5G. Although there are improvements going on in the existing technologies, companies like National Instruments (NI) are working on new ITU standards like 5G. 5G is in its early development stage and is anticipated to offer a significant measure of profit over the other existing technologies. It not only requires a software support but also requires a good hardware support for faster communication. New ways have been found to research, develop and test radios designed using 5G. The LabView Communication System Design Suite by NI makes this task possible in three steps: algorithm development by researchers; radio mapping using non-design tools by the designers for generating design standards; and radio implementation through code generation by the team.

"In order to support all these new technologies, the industry is working closely with product designers to help them write software to exploit new features, and working together on issues like power management to increase the efficiency of wireless communication devices without any compromise in the functionality," says Mrinmoy Purkayastha.

efits without any compromise in the system's reliability and measurement quality. Wireless sensors make the end product excitingly useful for the users.

The low-power ZigBee standards are basically optimised to the needs of the wireless sensor networks and thus offer self-healing mesh networking option along with robust self organising, low cost and complexity, the large extent of scalability and an excellent battery life. The smart sensors are also supporting wireless communication in smart meter and advanced meter infrastructures

(AMI), opening up a wide scope of designing options for the engineers.

Integrated transceivers

There are certain transceivers that not only offer a complete but also a high-performance mixed-signal system and RF on a single chip. In this category, there are some input/output RF based transceivers compatible with ZigBee standards that incorporate different layers of elements into another layer, facilitating a fast deployment feature.

This family of transceivers is

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